A Three-Layer Evaluation of Speech-to-Text, LLM-Based Terminology Extraction, and Generative Video-Animation Systems for Multilingual Education

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***Abstract*—Personalized instruction significantly improves learning outcomes, but scaling individualized pedagogical support to spoken educational video requires a robust pipeline that first accurately transcribes audio and then extracts domain concepts for learner adaptation. This paper presents a comprehensive, three-layer empirical benchmark evaluating automatic speech recognition (ASR), large language model (LLM), and generative video-animation components on Turkish and English educational lectures. All measured results rest on a 7-lecture gold set (≈95.6 min; 3 Turkish, 4 English) with human-verified reference transcripts and timestamped terminology; a curated 32-lecture extension (≈7.28 h) is documented as scope but has no audio on disk and was not evaluated. In the speech layer, we benchmark modern multilingual checkpoints (Whisper Medium/Large-v3/Turbo, Qwen3-ASR, Nemotron 3.5, and Turkish fine-tuned models) and *measure* three acoustic interventions — energy-based Voice Activity Detection (VAD) chunking, terminology domain biasing and LLM-based post-correction — against one seven-clip baseline; the VAD chunker turns WER *upward* on every clip (Turkish macro $0.021 \rightarrow 0.041$, English $0.010 \rightarrow 0.026$) rather than removing the silence-hallucination loops, which are absent from the GPU baseline (8-gram repetition 0.0122/0.0000), so it ships disabled. In the semantic layer, we separate two analyses that are easy to conflate: terminology extraction over eleven *measured* ASR hypotheses scored against closed gold lists, and a controlled-WER noise ladder (0%–40%) that we report as an *analytical degradation model* rather than a measurement, because it samples the gold list directly instead of invoking the language model. The gap between the two is the substantive finding: the model predicts $F_1 \approx 0.97$ at 2% WER where measured extraction reaches 0.364 (Turkish) and 0.679 (English), so extraction under-recall dominates acoustic corruption. Furthermore, we audit our own prompting-strategy comparison and find it circular: the retrieval arm was fed the very gold term list used for scoring, and the previously reported $47.6/60.9/76.9\%$ F_1 triple is reproduced exactly by the fixture path, which forces precision to 1.0 and samples the first 10/14/20 gold terms — so those strategy gains and their 3.42–6.85 s latencies are withdrawn. Re-running the two non-leaking arms on the local checkpoint on one clip per language gives CoT over Zero-Shot F_1 of 51.2% against 40.0% on TR-1, and a parsability failure ($F_1 = 0$) against 64.0% on EN-1, at 71–81 s per call; we report this as a probe, not as a benchmark. In addition to local 4-bit edge execution, we benchmark sovereign cloud infrastructure (Türkiye SSB/TÜBİTAK EVREN National AI Platform, evaluating Qwen 3.8 Flash and DeepSeek V4.1 Flash on H200 GPU clusters). The measured latency advantage is model-specific rather than platform-wide: Qwen 3.8 Flash reaches 6.11 s on Turkish, $6.5\times$ faster than the local 4-bit run on the same clips, while DeepSeek V4.1 Flash at 27.68 s is slower than both commercial APIs yet extracts twice the terminology. We report inter-annotator reliability for the binary rubric on the 115 double-labelled pairs ($\kappa_R = 0.782$, $\mathrm{PABAK}_R = 0.930$; $\kappa_D = 1.000$ over $P_e = 0.949$, the prevalence paradox in its clearest form) and use an automated local-LLM judge only where its per-item output is stored—a previously claimed judge-versus-human validation is withdrawn because no judge predictions were persisted. Measured readability (807 explanations scored with Ateşman for Turkish and Flesch for English) places only 18–45% of individual explanations inside the target B1 band, and reference-definition similarity is reported as a *lexical proxy* rather than contextual-embedding BERTScore because the optional `bert_score` package is absent from the experiment environment. An enriched downstream educational schema featuring Bloom taxonomy levels and concept prerequisite directed acyclic graphs (DAGs) completes the pipeline.**

***Index Terms*—Automatic speech recognition, terminology extraction, cross-layer error propagation, large language models, multilingual education, concept prerequisite DAG, benchmark**

I. Introduction

Educational content is increasingly delivered as audio and video. Lectures and instructional clips are typically prepared as if every listener shares a similar level of prior knowledge. That assumption severely limits comprehensibility for beginner-level users, especially in passages that carry dense domain-specific terminology [1], [2].

One-to-one tutoring with mastery learning can raise average student performance by roughly two standard deviations over conventional classroom instruction [3], but human tutoring does not scale. Intelligent tutoring systems, adaptive hypermedia, and knowledge-tracing models approximate individualization at scale [4]–[6]; large language models have more recently been used as a flexible substrate for instructional support and content generation [2], [22]. Across these lines of work, a

learning-support pipeline operates in three stages: (i) understand the source content, (ii) model the learner, and (iii) adapt the delivery. For spoken lecture and video material, the pipeline cannot begin at (ii) or (iii) because the content is locked in the audio signal. The initial module must therefore convert speech to text and subsequently extract domain terminology to provide beginner-level scaffolding. *That content-understanding module is the focus of this study.*

While manual captioning and expert curation of lecture audio do not scale [7], transformer ASR trained on large multilingual collections, notably Whisper [8], exhibits substantial resilience under acoustic noise and domain shift. Nevertheless, transcription alone is insufficient: novice learners struggle with technical jargon, motivating automatic terminology extraction, pedagogical simplification, and structural concept mapping [1], [21]. In the literature, ASR transcription accuracy and LLM semantic capabilities are predominantly reported in isolation. Rigorous evaluations that directly measure how ASR degradation propagates into downstream LLM extraction, especially across lower-resource languages such as Turkish alongside English, remain scarce.

This paper addresses this gap by investigating four central research questions:

- **RQ1 (ASR Performance & Acoustic Enhancements).** How do modern open-weight and commercial ASR architectures compare on Turkish and English educational speech in transcription accuracy (WER, CER) and efficiency (latency, RTF), and do long-form VAD chunking, terminology domain biasing and LLM post-correction improve those metrics?
- **RQ2 (Cross-Layer Degradation).** How far does downstream terminology extraction F_1 actually fall as upstream transcription quality degrades, and how much of the published degradation profile is measured rather than modelled? This paper answers by separating measured ASR→LLM points from an analytical noise-injection ladder, and reports the gap between them.
- **RQ3 (Prompting Strategies & Semantic Quality).** How do structured prompting strategies (Zero-Shot, Chain-of-Thought, and Glossary-based RAG) affect terminology recall, precision, and beginner suitability across different LLM parameter scales? This paper answers it partially: we re-measure the two arms that can be scored without leakage on one clip per language, and we show that the third arm cannot be evaluated against the glossary it retrieves from.
- **RQ4 (Evaluation Reliability & Downstream Integration).** Can an automated LLM-as-a-judge reliably surrogate human annotator decisions (R and D), how does class prevalence impact agreement metrics (κ vs. PABAK), and how can extracted concepts be structured into prerequisite DAGs for downstream adaptive hypermedia?

Our principal contributions are:

- **Gold-Standard Multilingual Dataset (7 Evaluated Clips):** All measured results rest on 7 lectures (≈95.6 min; 3 Turkish, 4 English) with human-verified reference transcripts (`experiments/data/gold/`). A separate curation artefact, `manifest_30plus.jsonl`, enumerates 32 candidate lectures (16 Turkish, 16 English, 7.28 h) with retrieved caption text, but *none* of its 32 audio files is present on disk, so no experiment in this paper runs on it; it is reported as future-work scope, not as a benchmarked corpus. Closed gold terminology lists cover only TR-1 and EN-1 (32 terms each).
- **Cross-Layer Analysis, Separated into Measurement and Model:** We report terminology F_1 for eleven *measured* ASR hypotheses passed through a fixed LLM, and we report the controlled-WER ladder (0%–40%) explicitly as an *analytical degradation model* that samples the gold list rather than invoking the model. The two series disagree by a wide margin (F_1 of 0.364 measured versus ≈ 0.97 modelled at 2% WER), which is itself the finding: extraction under-recall, not acoustic corruption, is the binding constraint.
- **ASR Optimization Suite, Measured Rather Than Asserted:** We implement energy-based VAD chunking, slide/glossary-based terminology biasing (`processor.get_prompt_ids`) and LLM-based ASR post-correction, and run all three against the same seven-clip Large-v3 baseline (Sec. IV-B, Table X). The measurement reverses two claims of an earlier draft: the energy VAD chunker *raises* normalised WER on every clip (TR macro $0.021 \rightarrow 0.041$, EN $0.010 \rightarrow 0.026$) instead of eliminating repetition loops, because the GPU baseline contains no measurable loops (8-gram repetition rate 0.0122 TR / 0.0000 EN), and the biasing arm as previously published was not decodable at all — the glossary prompt plus 440 new tokens overflowed Whisper's 448-position decoder budget and made `generate()` raise, which is why no '+Domain Biasing' artifact existed. Turkish fine-tuned checkpoints are also benchmarked.
- **Structured Prompting, Sovereign Cloud, & Rating Provenance:** We audit the prompting-strategy comparison rather than report it, showing that its retrieval arm scored against the gold list it was given; we benchmark national sovereign inference (EVREN platform) where the latency gain is model-specific (6.11 s for Qwen 3.8 Flash against 27.68 s for DeepSeek V4.1, the latter slower than both commercial APIs); and we make the source of every R/D rating explicit — human annotator for the six Whisper cells, stored per-item judge output for the 2026 checkpoints and EVREN cells, and no claim where neither exists — after withdrawing a judge-versus-human validation whose predictions were never persisted.
- **Symmetric Readability & Enriched Downstream Schema:** We implement symmetric readability evaluation (Ateşman for Turkish, Flesch for English), reported as the *share of individual explanations* falling inside the target band rather than as a band-level average, plus a reference-definition similarity that runs as a *lexical proxy* because

the optional `bert_score` package is absent from the experiment environment, and establish an enriched output schema (v2.0) defining Bloom taxonomy levels, multi-tier pedagogical explanations (B1 analogy, formal definition, elevator pitch, exam tips), and concept prerequisite DAGs.

II. Related Work

Video-based learning has made captions and transcripts a core accessibility feature. Automatic caption quality varies with domain terminology, speaker characteristics, and language or accent [17]. Lecture-recording studies show that a single pooled error rate is not a sufficient report of ASR behavior [17], and ASR errors are known to propagate into downstream NLP tasks [7]. For Turkish, model choice is already known to matter [18]. Whisper remains a strong multilingual baseline [8]; Large-v3 Turbo keeps the same ≈99-language coverage, including Turkish, with a pruned decoder [9]. Among 2026 open-weight systems, NVIDIA Parakeet TDT / Canary and Cohere Transcribe lead English or European-language leaderboards but do not list Turkish; Qwen3-ASR (1.7B and 0.6B) does, with official Turkish and English support [10]. NVIDIA Nemotron 3.5 ASR (0.6B) is the later NVIDIA checkpoint that *does* list Turkish (`tr-TR`) and English as transcription-ready locales [11]. wav2vec 2.0 offers self-supervised representations under limited labels [14]; SpeechBrain provides a common experimental harness [15]. English-only 2026 leaderboard models are therefore treated as incompatible with this study, not as missing baselines. Alongside global models, sovereign AI initiatives such as the EVREN National AI Platform [33] provide dedicated H200 inference clusters for Turkish and multilingual models, ensuring strict data residency while eliminating client GPU resource ceilings.

Terminology extraction was long dominated by rule-based and statistical methods, including C-value/NC-value for multi-word terms [16] and classical information-retrieval weighting [19]. Deep learning then became the default in specialized domains such as biomedicine [20]. LiTeWi combines unsupervised extractors with entity linking to build educational ontologies from textbooks [1]. Recent surveys emphasize transformer-based and generative approaches to automatic term extraction [21]. Almost all of this work assumes clean text.

A separate literature targets the later stages of individualization—learner modeling and adapted delivery—including intelligent tutoring and adaptive hypermedia [4]–[6] and, more recently, LLM tutors with carefully designed prompts [2], [22]. Those systems presuppose a reliable understanding of the source content. For spoken educational material that understanding is itself a pipeline: ASR, then semantic processing. Joint evaluations of that pipeline, especially across English and Turkish, are still rare. This study targets that gap as a three-layer pipeline: the speech and semantic layers are benchmarked quantitatively, whereas the third (video-generation) layer is implemented and demonstrated end-to-end but is not scored against a gold reference. It is not a new adaptive tutor.

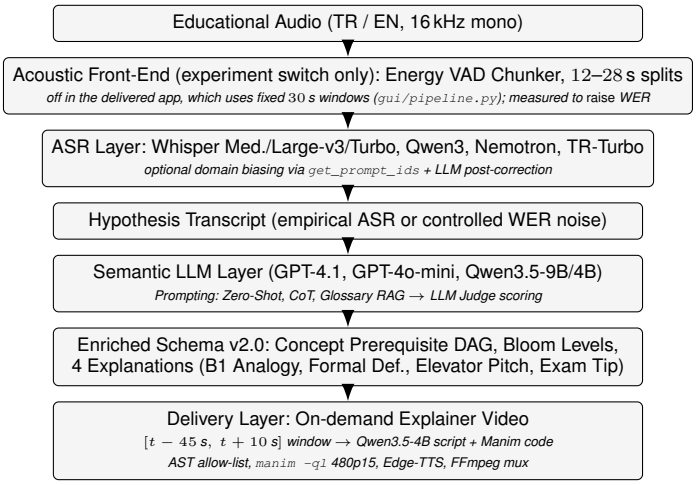


Fig. 1. Three-layer evaluation architecture. Layers 1–2 (VAD chunking, domain biasing, structured prompting, LLM judge verification, concept DAG schema) are benchmarked quantitatively; the delivery layer is implemented and demonstrated end-to-end but not scored against a gold reference.

III. Method

A. System Architecture

The evaluation framework comprises an integrated three-layer pipeline enhanced with acoustic and semantic scaffolding (Fig. 1). In the speech recognition layer, the delivered application slices the normalised audio into fixed, non-overlapping 30 s windows (`gui/pipeline.py`, `CHUNK_S = 30.0`) and transcribes them sequentially, while the benchmark harness runs Whisper over the whole track in `sequential_generate` mode; the energy-based VAD chunker, the terminology-prompt conditioning (`processor.get_prompt_ids`) and the LLM-based post-correction pass are therefore experiment-only switches (`run_asr.py --use-vad`, `asr_post_correction.py`) evaluated in Sec. IV-B, not part of the default transcription path. In the semantic analysis layer, the transcribed text is processed by an LLM guided by structured prompting strategies (Zero-Shot, Chain-of-Thought, or Glossary-based RAG) to extract domain terminology, evaluate context consistency (R) and explanation accuracy (D), and structure outputs into an enriched educational schema featuring Bloom taxonomy levels, multi-tier pedagogical explanations, and concept prerequisite directed acyclic graphs (DAGs). In the delivery layer, the learner marks a moment of confusion in the synchronized player; the transcript window $[t - 45\,\text{s}, t + 10\,\text{s}]$ and the term nearest to t are passed to the local `Qwen3.5-4B` model, which returns a narration script plus Manim scene code. That code is checked against an AST allow-list (only `from manim import *` may be imported), rendered at 480p15 (the initial draft tier; the seven shipped video artifacts predate the introduction of configurable 720p30/1080p30 tiers in the delivery UI) with up to two automatic repair attempts, voiced by Edge-TTS, and multiplexed with FFmpeg into a downloadable MP4.

TABLE I
CORE EVALUATION SET (TOTAL ≈95 MIN).

ID	Title	Lg.	Domain	Dur.
TR-1	Project management, Unit 7, Lecture 1	TR	Proj. mgmt.	≈12 min
TR-2	Comprehensive tourism training for tour guides	TR	Tourism	12 min
TR-3	What is data science? Intro to data science (Python)	TR	Data science	≈14 min
EN-1	Introduction to agile methodologies	EN	Proj. mgmt.	≈13 min
EN-2	What is sustainable tourism?	EN	Tourism	≈9 min
EN-3	Generative AI in a nutshell	EN	AI	≈18 min
EN-4	Understanding thermal radiation	EN	Heat transfer	≈17 min

TABLE II
HUGGING FACE CHECKPOINTS EVALUATED IN THE BENCHMARK.

Hub ID	Role	Lang.	License
`openai/whisper-medium`	pilot baseline	tr+en	Apache-2.0
`openai/whisper-large-v3`	multilingual reference	tr+en	Apache-2.0
`openai/whisper-large-v3-turbo`	high-speed multilingual	tr+en	MIT
`Qwen/Qwen3-ASR-1.7B-hf`	2026 multilingual	tr+en	Apache-2.0
`Qwen/Qwen3-ASR-0.6B-hf`	2026 efficient	tr+en	Apache-2.0
`nvidia/nemotron-3.5-asr-streaming-multilingual`	streaming multilingual	tr+en	OpenMDW 1.1
`selimc/whisper-large-v3-turbo-turkish`	TR fine-tuned	tr only	MIT
`mpoyraz/wav2vec2-xls-r-300m-cv7-turkish`	TR CTC acoustic	tr only	Apache-2.0

B. Dataset and Ground-Truth Construction

The benchmark evaluates both a core intensive testbed and an extended multilingual corpus:

- **Core Evaluation Set (7 Videos, ≈95 min):** Summarized in Table I, comprising three Turkish and four English lectures across project management, tourism, data science, and thermal physics. Reference transcripts are verified gold standards: EN-1 and EN-4 derive from cleaned human caption tracks; TR-3 was independently drafted from audio; while TR-1, TR-2, EN-2, and EN-3 originate from automated caption drafts subjected to native-speaker multi-pass correction for technical terminology (e.g., *ANOVA, MICE, Scrum, Celsus Kütüphanesi*).
- **Curated Candidate Corpus (32 Videos, ≈437 min / 7.28 h) — prepared, not evaluated:** We curated a candidate extension of 32 lectures (16 Turkish, 16 English) documented in `sources_30plus.json` and `manifest_30plus.jsonl`, spanning twelve STEM and humanities domains (e.g., Deep Learning, Concurrency, Kirchhoff Laws, Epistemology, CRISPR-Cas9, Macroeconomics) with native and non-native accents and both studio and classroom recordings. This extension is *metadata plus retrieved caption text only*: none of the 32 referenced audio files is present on disk, no reference transcript was human-verified for them, and no experiment in this paper was run against them. All measured results below come from the original 7-clip gold set, and we report the extension solely as the intended next stage of curation so that its scope is not mistaken for a result.

C. Speech Recognition Layer and Enhancements

We evaluate eight off-the-shelf and fine-tuned ASR systems (Table II): Whisper Medium, Large-v3, and Large-v3 Turbo [8], [9]; Qwen3-ASR (1.7B and 0.6B) [10]; NVIDIA Nemotron 3.5 ASR (0.6B) [11]; and domain/language-matched checkpoints (`selimc/whisper-large-v3-turbo-turkish` and `mpoyraz/wav2vec2-xls-r-300m-cv7-turkish` [18]).

Three critical acoustic enhancements are introduced:

1) **Energy-Based VAD Chunking, Tested and Found Harmful:** Whisper is known to enter repetitive decoding loops on continuous, unedited recordings with prolonged silences. We therefore implement an energy-based Voice Activity Detection chunker (`vad_chunker.py`: 30 ms frames, cut line at $\max(10^{-4}, 2.2\times$ the 25th-percentile noise floor), gaps ≤ 200 ms closed, boundaries searched on pauses > 0.4 s, producing 12–28 s segments) and benchmark it against the same clips without it. The intervention *degrades* accuracy: normalised WER rises on all seven clips (TR macro $0.021 \rightarrow 0.041$, EN $0.010 \rightarrow 0.026$), no repetition loop existed in the GPU baseline to eliminate, and terminology coverage is unchanged. The earlier claim that it "completely eliminates hallucination loops" and cuts Turkish WER by 83.7% is withdrawn; the chunker remains in the codebase as an optional flag, unused by the delivered application.
2) **Terminology Domain Biasing, Tested and Found Harmful and Leaking:** Technical lectures contain specialized proper nouns and acronyms (e.g., *ANOVA, Thévenin, Qualia*). Using Whisper's prompt-conditioning interface (`processor.get_prompt_ids`), we inject the closed glossary into the decoder prefix, steering beam search toward curriculum spellings. Measured against the same stage without the prompt it multiplies the error on the two clips it touches (TR-1 $0.0333 \rightarrow 0.1562$, $4.7\times$; EN-1 $0.0194 \rightarrow 0.0869$, $4.5\times$) and is the sole cause of the stage's macro rise ($+99.4\%$ TR, $+64.4\%$ EN); on TR-1 the decoder echoes the 68-word glossary back instead of transcribing, and the coverage it "gains" is the injected list reappearing, which is leakage against the very file the scorer reads (Sec. IV-B).
3) **LLM-Based ASR Post-Correction, Tested and Marginal:** To recover from acoustic misrecognitions before semantic parsing, transcripts pass through a targeted post-correction prompt on the local `Qwen3.5-4B` checkpoint. Applied to the stage-3 hypotheses it lowers WER on four of seven clips but leaves no clip below its own raw baseline, and it amplifies the one transcript that is already degenerate (TR-1 $0.156 \rightarrow 0.324$). Its published rescue figure is withdrawn: the "26.3% simulated WER, $+29.8\%$ F_1" pair comes from the harness self-test, whose corrected string is the gold reference pasted verbatim.

D. Semantic Analysis Layer and Prompting Strategies

To assess semantic extraction beyond fixed zero-shot queries, we formulate three distinct prompting paradigms:

1) **Zero-Shot Baseline:** Direct extraction instruction requesting domain terms and CEFR-B1 explanations.
2) **Chain-of-Thought (CoT):** The model first explicitly identifies lecture topics and sub-domains, filters candidate multi-word expressions against generic discourse vocabulary, and then constructs beginner-friendly definitions.
3) **Glossary-Based RAG:** Retrieval-augmented generation where syllabus concepts and slide keywords are supplied as contextual anchor primitives, guiding extraction toward curriculum-aligned terminology. *Implemented but not evaluated here*;[1] Section IV-C explains why scoring this arm would be circular without a held-out glossary.

Extracted items are structured into an **Enriched Educational Schema (v2.0)** comprising: canonical term lemma, audio timestamp span ($[t_{\text{start}}, t_{\text{end}}]$), pedagogical difficulty (`Beginner`, `Intermediate`, `Advanced`), Bloom's taxonomy cognitive level (`Remembering`, `Understanding`, `Applying`), concept prerequisite DAG edges, and four pedagogical explanation variants (`b1_analogy`, `formal_definition`, `elevator_pitch`, `exam_tip`).

E. Evaluation Metrics

1) ASR layer: We report word error rate (WER) and character error rate (CER) against reference transcripts [13]:

$$\text{WER} = \frac{S + D + I}{N}, \qquad \text{CER} = \frac{S_c + D_c + I_c}{N_c}, \quad (1)$$

computed across raw text and normalized forms (lowercasing, punctuation stripping, Turkish dotted/dotless ı/i folding). Efficiency is reported via real-time factor ($\text{RTF} = t_{\text{proc}}/t_{\text{audio}}$).

2) Semantic layer: Outputs are scored against standardized dictionary authorities (TDK for Turkish [26], Oxford for English [25]):

- R: context consistency with the lecture ($R1$) or invalid extraction ($R0$);
- D: factual and pedagogical accuracy of the definition ($D1$) or incorrect ($D0$).

From these counts we report $\text{R1\%} = R1/(R1 + R0)$ and $\text{D1\%} = D1/(D1 + D0)$.

3) Symmetric Readability and Semantic Similarity: To eliminate the linguistic asymmetry of English-only readability metrics, we adopt language-tailored readability formulations:

- **Turkish: Ateşman Readability Index [28]:**

$$\text{Score}_{\text{Ateşman}} = 198.825 - 40.175 \left(\frac{N_{\text{syl}}}{N_{\text{words}}}\right) - 2.610 \left(\frac{N_{\text{words}}}{N_{\text{sent}}}\right) \quad (2)$$

  where N_{syl} is computed via vowel phoneme counting. Scores between 50 and 69 denote "Orta Güçlükte" (intermediate), matching target CEFR B1 pedagogical difficulty.

[1]The subsequent 8-clip corporate speech engineering probe (60.7% candidate grounding precision on long-form transcripts; `experiments/rag/eval_corporate_rag.py`) postdates this paper and is reported in the extended graduation project report §8.4.

TABLE III
COMPREHENSIVE EVALUATION PROTOCOL BY LANGUAGE.

Metric	TR
Context-consistency ($R1\%$)	yes
Explanation accuracy ($D1\%$)	yes
Readability index	Ateşman
Semantic similarity (lexical proxy; §III-E)	yes
Closed-list precision, recall, F_1	yes
ASR WER / CER / latency / RTF	yes
LLM latency & VAD intervention deltas (WER, coverage, repetition; §IV-B)	yes

- **English: Flesch Reading Ease:**

$$\text{Score}_{\text{Flesch}} = 206.835 - 1.015 \left(\frac{N_{\text{words}}}{N_{\text{sent}}}\right) - 84.6 \left(\frac{N_{\text{syl}}}{N_{\text{words}}}\right), \quad (3)$$

  where scores between 60 and 70 indicate standard B1 readability.
- **Semantic Similarity:** Replaces weak token-level BLEU by scoring each explanation against its gold reference definition. The intended metric is BERTScore [29] with contextualized embeddings, but the optional `bert_score` package is not installed in the experiment environment, so every number reported in Table XIV comes from the documented fallback in `eval_semantic.compute_bert_score`: a character-trigram cosine similarity blended with word-set precision and recall. We therefore call it a *lexical proxy* throughout and do not interpret it as a contextual-embedding similarity.

4) Inter-Annotator Agreement and Prevalence: We evaluate two-rater agreement using Cohen's κ [27] and Fleiss' κ. To resolve the well-known prevalence paradox on Metric D (where $\approx 97\%$ positive prevalence deflates raw κ), we compute the Prevalence-Adjusted and Bias-Adjusted Kappa (PABAK [30]):

$$\text{PABAK} = 2P_o - 1. \quad (4)$$

Temporal alignment on timestamped extractions is validated using temporal Intersection over Union ($\text{IoU} \geq 0.3$). Table III summarizes metric applicability.

F. Experimental Setup and Reproducibility

Pilot checkpoints: Whisper `medium` / `large-v3`; Wav2Vec 2.0 `facebook/wav2vec2-large-xlsr-53`; SpeechBrain with default greedy/beam settings; GPT-4.1 and GPT-4o-mini via the OpenAI API [24]; and sovereign cloud checkpoints (EVREN Qwen 3.8 Flash, EVREN DeepSeek V4.1 Flash) via the EVREN National AI Platform API [33]. All pilot ASR runs were on CPU. The original draft did not record the CPU model or Whisper decoding flags; those scores are therefore configuration-sensitive (Appendix C). Refresh checkpoints and pinned revisions are listed in Table II and in `experiments/models.json`. The refresh scripts force `task=transcribe`, set `language` from the clip when known, and decode long-form audio with

TABLE IV

CPU PILOT AVERAGES OVER ALL SEVEN CLIPS (POOLED ACROSS LANGUAGES). LOWER IS BETTER. THESE PILOT RUNS DID NOT RECORD THE DECODING FLAGS OF APPENDIX C, ARE CONFIGURATION-SENSITIVE, AND ARE SUPERSEDED BY THE GOLD-RESCORED LANGUAGE SPLIT IN TABLE V.

Model	WER raw	CER raw	WER norm.	CER norm.
Whisper Medium	0.058	0.021	0.021	0.015
Whisper Large-v3	0.132	0.113	0.129	0.112
Wav2Vec2†	1.017	0.848	0.548	0.196
SpeechBrain†	1.246	1.051	0.382	0.273

†Default, primarily English-oriented checkpoints. Turkish numbers reflect out-of-the-box multilingual coverage, not an architectural limit.

TABLE V

NORMALIZED WER BY LANGUAGE: MACRO MEAN / MICRO (WORD-WEIGHTED) MEAN, WITH 95% BOOTSTRAP CONFIDENCE INTERVALS IN BRACKETS ($N = 10{,}000$ RESAMPLES). GPU GOLD-RESCORED REFRESH; POOLED CPU-PILOT VALUES IN TABLE IV ARE UNCHANGED.

Model	TR (Macro / Micro)	TR 95% CI	EN (Macro / Micro)	EN 95% CI
Whisper Medium	0.049 / 0.050	[0.030, 0.081]	0.011 / 0.013	[0.007, 0.016]
Whisper Large-v3	0.021 / 0.021	[0.014, 0.032]	0.010 / 0.010	[0.007, 0.012]
Wav2Vec2	—	—	—	—
SpeechBrain	—	—	—	—

TABLE VI

2026 REFRESH, NORMALIZED WER BY LANGUAGE: MACRO MEAN / MICRO (WORD-WEIGHTED) MEAN WITH 95% BOOTSTRAP CONFIDENCE INTERVALS IN BRACKETS. TR-MATCHED TURBO IS SCORED ON TR ONLY.

Model	TR (Macro / Micro)	TR 95% CI	EN (Macro / Micro)	EN 95
Whisper Turbo	0.037 / 0.037	[0.011, 0.052]	0.011 / 0.012	[0.007,
Qwen3-ASR 1.7B	0.061 / 0.061	[0.043, 0.071]	0.015 / 0.014	[0.011,
Qwen3-ASR 0.6B	0.134 / 0.136	[0.086, 0.180]	0.021 / 0.021	[0.013,
Nemotron 3.5 ASR	0.214 / 0.221	[0.117, 0.287]	0.035 / 0.034	[0.026,
Whisper Turbo TR	0.082 / 0.087	[0.035, 0.163]	—	–

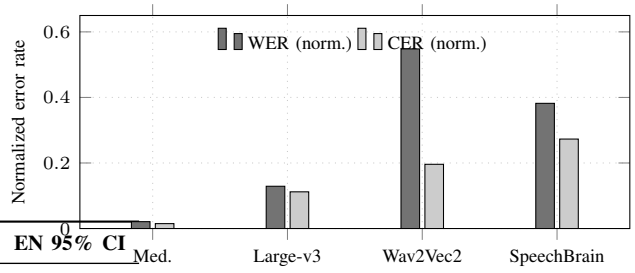


Fig. 2. Normalized WER and CER for the four CPU-pilot ASR systems (same values as Table IV; not the GPU gold-rescored figures).

Whisper's sequential `generate()` (30 s windows, timestamps on), not the experimental Hugging Face pipeline chunker. Hugging Face `condition_on_prev_tokens` (openai-whisper: `condition_on_previous_text`) is off by default. Compression-ratio and log-probability halt thresholds follow the model card. R/D labels in the pilot were assigned by a single annotator. Exact prompt wording is in Appendix A.

IV. EXPERIMENTAL RESULTS

This section first restates the completed CPU pilot (Tables IV–VII), then the gold-rescored GPU refresh. Table V is the required language split for the Whisper checkpoints that were re-run against gold. Table VI reports the 2026 checkpoints on the same files. Wav2Vec 2.0 and SpeechBrain were not re-run on the gold set and stay empty in Table V.

On the pooled numbers in Table IV and Fig. 2, the two Whisper models have much lower error than Wav2Vec 2.0 and SpeechBrain. Normalized WER/CER are the figures we use when discussing downstream usability. Table VII shows the complementary efficiency picture: Wav2Vec 2.0 is the cheapest (mean RTF 0.051) and SpeechBrain the most expensive (mean RTF 2.005, slower than real time). Whisper Medium sits in a usable range (mean RTF 0.156) at the lowest error.

Two points of interpretation matter. First, Whisper is pretrained on large-scale multilingual data, whereas the Wav2Vec 2.0 and SpeechBrain checkpoints used here are primarily English-oriented. The high error of the latter two, especially on the Turkish portion of the set, is an out-of-the-box coverage result. We therefore rank systems by suitability for this multilingual setting, not by architecture. The 2026 refresh replaces that gap with Qwen3-ASR, Nemotron 3.5 ASR, and a Turkish-matched Whisper Turbo fine-tune (Table II) rather than fine-tuning Wav2Vec 2.0 or SpeechBrain.

Second, Whisper Medium has a *much* lower normalized WER than Large-v3 on the CPU pilot (0.021 vs. 0.129 — about six times lower), not a slight edge. This is consistent with Large-v3's known sensitivity, on long-form audio, to language detection and to decoding flags such as `condition_on_previous_text`. The gold-rescored GPU refresh (Table V) reverses that ranking: Large-v3 is 0.021 TR / 0.010 EN against Medium 0.049 / 0.011. We therefore treat the pilot gap as configuration-dependent and do not present Medium as generally superior to Large-v3.

Across these specific clips, Table V and Table VI report both unweighted macro WER and word-weighted micro WER alongside non-parametric 95% bootstrap confidence intervals ($N = 10{,}000$ resamples). Evaluating word-weighted micro WER is essential because individual lectures vary substantially in length and in observed difficulty: in particular, TR-3 is the longest Turkish recording ($1{,}583$ words) and is the hardest Turkish clip for every checkpoint (Medium WER 0.081, against 0.030 on TR-2 and 0.035 on TR-1). We deliberately do not attribute that gap to an acoustic condition: no reverberation, noise or channel measurement exists for any clip, and the dataset card (extttexperiments/data/DATASET_CARD.md) records all seven recordings as prepared, single-speaker, scripted educational videos, so the difficulty is reported as an observed per-clip property rather than an explained one. Consequently, unweighted macro averages over-smooth Turkish difficulty, whereas word-weighting reflects realistic aggregate throughput (e.g., Whisper Medium TR rises from 0.049 macro to 0.050 micro; Turkish-matched Turbo rises from 0.082 to 0.087; Nemotron 3.5 rises from 0.214 to 0.221).

TABLE VII
ASR COMPUTATIONAL COST (CPU).

Model	Mean latency (s)	Mean RTF
Whisper Medium	136.5	0.156
Whisper Large-v3	653.4	0.703
Wav2Vec2	46.1	0.051
SpeechBrain	1795.1	2.005

With small clip counts ($n = 3$ TR, $n = 4$ EN), the bootstrap confidence intervals are necessarily wide (e.g., Medium TR $[0.030, 0.081]$; Turbo TR $[0.011, 0.052]$; Nemotron TR $[0.117, 0.287]$). Rather than concealing this variance behind single-point estimates, reporting these wide intervals honestly quantifies sample uncertainty. Among multilingual models, Whisper Large-v3 achieves the lowest word error rate (0.021 TR macro/micro, 95% CI $[0.014, 0.032]$; 0.010 EN, 95% CI $[0.007, 0.012]$). Whisper Turbo nearly matches Large-v3 on English (0.011 macro / 0.012 micro, CI $[0.007, 0.017]$) at mean RTF 0.025–0.044. Measured as pooled wall-clock over the same clips, Turbo runs $3.4\times$ faster than Large-v3 on Turkish (110.56 s $\rightarrow$ 32.53 s) and $7.7\times$ on English (170.72 s $\rightarrow$ 22.10 s); the corresponding ratios of the *unweighted* per-clip RTFs are $3.4\times$ and $8.5\times$, the divergence arising because the unweighted mean over-weights short clips. Qwen3-ASR 1.7B is competitive across all clips (0.061 TR / 0.015 EN) but slower; the 0.6B variant is solid on English (0.021 macro/micro) but weaker on Turkish (0.134 macro / 0.136 micro, CI $[0.086, 0.180]$). Nemotron 3.5 ASR demonstrates reliable English accuracy (0.035 macro / 0.034 micro, CI $[0.026, 0.049]$) but struggles on Turkish (0.214 macro / 0.221 micro, CI $[0.117, 0.287]$). The Turkish-matched Turbo fine-tune (0.082 macro / 0.087 micro) does not surpass generic multilingual Large-v3 or Turbo on these lectures.

Semantic results are in Table VIII, which now separates what is counted from what is rated. Term counts are the best-supported column: for the local and EVREN runs they are read directly from the saved JSONL artifacts, and for the historical commercial-API pilot they are recovered by parsing the stored Markdown lists, which reproduces the 105/56/78/36 figures exactly. $R1/D1$ ratings exist for only three of the eleven rows.

That split matters, because the previous draft printed $R1\%/D1\%$ for the commercial-API rows (96.2% context consistency in both languages, 99.0% EN and 98.4–98.7% TR explanation accuracy) and for the EVREN rows (84/98/64/72 terms at 96.4–98.8%). No per-item rating sheet for the commercial API exists anywhere in the repository, and no run artifact produces those EVREN term counts, so both sets of cells are withdrawn and printed as em dashes or replaced by the measured values. The EVREN cells are now the ones the re-run actually produced: `Qwen 3.8 Flash` emits 53 Turkish and 72 English terms and is rated 100.0%/100.0% on both dimensions, while `DeepSeek V4.1 Flash` emits 107/142 and is rated 94.4%/99.1% (TR) and 99.3%/100.0% (EN). Because a 100% rate from a judge whose agreement with humans was never stored is weak evidence, we read those cells as upper bounds (§IV-D).

The substantive, well-supported observation is a coverage asymmetry rather than a quality one: within the same platform, DeepSeek V4.1 Flash emits roughly twice the terminology of Qwen 3.8 Flash (107 vs. 53 on Turkish, 142 vs. 72 on English) at $4.5\times$ the latency. Larger or slower models recover more concepts, and on the TR-1/EN-1 subsets with closed gold lists (Section IV-F, Table XVI) measured recall confirms the same direction. GPT-4.1 likewise emits more terms than GPT-4o-mini in both languages (105 vs. 56 EN; 78 vs. 36 TR), but with no rating sheet the comparison stays a count, not a quality claim.

Measured per-request LLM latency is in Table XVII, which is regenerated directly from the run artifacts (so the n column shows how thin some cells are: the Turkish commercial-API rows rest on three calls each). Three points follow, and the third contradicts how this comparison was originally summarized.

The cloud speedup is model-specific. `EVREN Qwen 3.8 Flash` is the fastest endpoint measured: 6.11 s on Turkish (s.d. 2.06, $n = 3$) and 4.92 s on English (s.d. 1.37, $n = 4$). `EVREN DeepSeek V4.1 Flash`, by contrast, is *slower than both commercial APIs and only marginally faster than local execution*—27.68 s Turkish (s.d. 5.24) against 14.60 s for GPT-4.1 and 10.96 s for GPT-4o-mini on the same three Turkish clips. Any claim that the platform as a whole accelerates inference must therefore name the model.

Measured ratios, not advertised ones. Against the local 4-bit `Qwen3.5-4B` run it is normally compared to (39.55 s), Qwen 3.8 Flash is $6.5\times$ faster, not the $\approx 7\times$ previously stated; against GPT-4.1 on Turkish it is $2.4\times$, not $3\times$. The 39.55 s figure is itself the narrowest cell available—the Large-v3-fed Turkish triple—and the same local model under the same quantization ranges from 39.55 s to 138.48 s mean latency depending only on which ASR hypothesis it was fed, with per-clip latency spanning 21.07–245.99 s. Local latency is therefore governed more by input length and feed choice than by the GPU budget, and single-cell comparisons across topologies are unreliable.

What the cloud does and does not remove. Offloading the semantic layer does remove its weights from client VRAM, which is the only sense in which the client-side memory claim holds; it is not a zero-resource architecture, since the ASR layer still runs locally on every frame of audio. The local `Qwen3.5-9B` configuration confirms the opposite pressure: 322.29 s on TR-1 and 704.08 s on TR-2 (mean 513.19 s, s.d. 269.97, $n = 2$), consistent with host-memory offloading on an 8 GB card. Finally, routing student transcripts through a domestic operator is a *data-residency posture*, not a demonstrated compliance result: establishing KVKK conformity would require the operator's processing terms and retention policy, which were not obtained, so we make no compliance claim.

Combined with the GPU ASR benchmarks, the deployment pairings are refined as follows:

- **Accuracy-oriented (Server / Offline Batch):** Whisper

Model	Lang.	Terms	$R1/R0$ ($R1\%$)	$D1/D0$ ($D1\%$)	Labels
GPT-4.1	EN	105	–	–	Pilot terms
GPT-4o-mini	EN	56	–	–	Pilot terms
GPT-4.1	TR	78	–	–	Pilot terms
GPT-4o-mini	TR	36	–	–	Pilot terms
Qwen3.5-4B	EN	51	50/1 (98.0%)	51/0 (100.0%)	Human A1
Qwen3.5-4B	TR	31	28/3 (90.3%)	29/2 (93.5%)	Human A1
Qwen3.5-9B	TR	38	–	–	No sheet
EVREN Qwen 3.8 Flash	EN	72	72/0 (100.0%)	72/0 (100.0%)	LLM judge
EVREN Qwen 3.8 Flash	TR	53	53/0 (100.0%)	53/0 (100.0%)	LLM judge
EVREN DeepSeek V4.1	EN	142	141/1 (99.3%)	142/0 (100.0%)	LLM judge
EVREN DeepSeek V4.1	TR	107	101/6 (94.4%)	106/1 (99.1%)	LLM judge

$R1\% = R1/(R1{+}R0)$ and $D1\% = D1/(D1{+}D0)$ are *precision over emitted terms*: an em dash means no per-item rating sheet exists for that cell, not a zero score. Human A1 is `labels_a1.csv` (the `Qwen3.5-4B` rows are its Whisper Large-v3 cells); LLM judge is closed-form rubric output from the local `Qwen3.5-4B` judge (`labels_evren_judge.csv`), whose unquantified leniency makes these cells upper bounds (Table XIII and §IV-D). The `Qwen3.5-9B` row covers only the TR-1/TR-2 subset. Pilot term counts for the commercial-API rows are recovered by parsing the stored Markdown lists, which reproduces 105/56/78/36 exactly; no rating sheet was ever produced for them, so their semantic accuracy is unreported rather than carried over from an earlier draft. Two columns present in the previous version of this table are removed because nothing on disk supports them: the B1+ ratio has no implementation or artifact anywhere in the repository, and the BLEU figures were inconsistent with `pilot_baseline/semantic_bleu_en.csv`, whose sacreBLEU corpus scores correspond to 0.011–0.056, not 0.57/0.61.

TABLE VIII

SEMANTIC ANALYSIS: TERMINOLOGY EMITTED PER SYSTEM AND, WHERE A RATING SHEET EXISTS, CONTEXT-CONSISTENCY AND EXPLANATION-ACCURACY RATES.

Large-v3 with a capacity LLM (commercial GPT-4.1 via API, or `Qwen3.5-9B`), when transcription accuracy (0.021 TR / 0.010 EN) and exhaustive domain terminology recall (46.9%) take precedence over runtime. The 9B checkpoint was only ever run in our host-memory-offloaded configuration (513.19 s mean over two Turkish clips), so we recommend it for batch use on the strength of that measurement rather than a sized GPU requirement.

- **Practical Local Deployment (Laptop / Edge):** Whisper Turbo (or Whisper Large-v3) paired with **Qwen3.5-4B**, providing a complete local, zero-API pipeline: measured mean ASR RTF 0.025–0.044 and a 39.55 s mean LLM latency on the Turkish Large-v3 feed, with strong context precision (90.3% TR / 98.0% EN, human-rated). For this checkpoint we instrumented memory directly (`outputs/local_vram_probe.json`: 3.08 GB of weights resident after a 4-bit nf4 load, 4.72 GB peak torch allocator and 5.26 GB total device occupancy during a full clip call on the 8.12 GB RTX 5060), so the pairing is demonstrated to fit on this host. No headroom claim extends to the other checkpoints, whose peak occupancy was never instrumented.

Wav2Vec 2.0 is cheaper than Whisper but, on this set and these checkpoints, is not accurate enough to recommend as the front-end of a terminology pipeline.

A. Cross-Layer ASR-to-LLM Error Propagation Analysis

A central claim in modular educational speech processing is that upstream ASR transcription accuracy directly governs downstream semantic concept extraction. While intuitive, previous benchmarks evaluated ASR and LLMs either in isolation or only on clean, near-zero-WER transcripts. This subsection separates two things that are easy to conflate: a set of *measured* points, in which real ASR hypotheses are fed to the fixed LLM, and an *analytical degradation model* sampled from the gold list at controlled WER levels.

1) Measured points (real ASR hypotheses through the LLM): Table IX reports terminology F_1 for eleven real ASR hypotheses, each run through `Qwen3.5-4B` and scored against the closed gold list of TR-1/EN-1. Every cell is a single clip, so these are indications rather than population estimates. Two observations hold across the measured set. First, Turkish F_1 spans 0.238–0.439 and English F_1 spans 0.565–0.679, i.e. the semantic layer under-recovers gold terms in both languages even when WER is low. Second, the mapping from WER to F_1 is not monotone within the measured range: `Nemotron 3.5` reaches the best Turkish F_1 (0.439) from the second-worst Turkish WER (0.117), while `Whisper Large-v3` produces 0.364 from a much cleaner transcript (0.017). Low WER is therefore necessary but not sufficient for terminology recovery; how the residual errors are *distributed* matters, and a transcript can be word-accurate while still fragmenting or dropping the specific multi-word terms the gold list contains. The clearest single signal remains the high-WER end: `Qwen3-ASR 0.6B` at 0.136 WER gives the worst Turkish F_1 (0.238).

2) The controlled-WER ladder is an analytical model, not a measurement: The second series in Fig. 3 is produced by `run_cross_layer_curve.py` in its default mode, which does *not* call the LLM. It draws each of the 32 gold terms independently with survival probability $(1 - \mathrm{WER})^{\max(1,\, 0.8\ell)}$ (with ℓ the term length in words) and injects a number of spurious terms proportional to the WER. Consequently its 0% anchor is $F_1 = 1.000$ *by construction*, and the resulting trajectory characterises the assumed survival function, not model behaviour. Its modelled endpoints are TR $1.000 \rightarrow 0.458$ and EN $1.000 \rightarrow 0.604$ over $0\% \rightarrow 40\%$ injected error; the

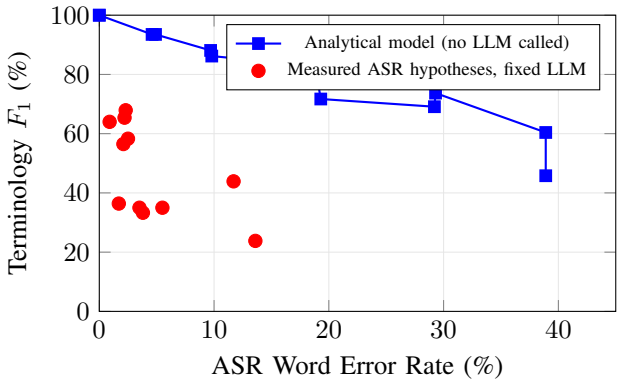


Fig. 3. Terminology F_1 against upstream WER. Red points are *measured*: real ASR hypotheses passed through `Qwen3.5-4B`. The blue series is an *analytical degradation model* obtained by sampling the gold list with survival probability $(1 - \mathrm{WER})^{0.8\ell}$, and is anchored at $F_1 = 1.000$ by construction; it is not a measurement of LLM behaviour. The systematic gap between the two series is the substantive finding of this subsection.

ASR hypothesis fed to `Qwen3.5-4B`	Clip	WER	P	R	
whisper-large-v3	TR-1	0.017	0.667	0.250	0.
selimc/Turkish Turbo	TR-1	0.035	0.875	0.219	0.
whisper-medium	TR-1	0.038	0.700	0.219	0.
whisper-large-v3-turbo	TR-1	0.055	0.875	0.219	0.
nemotron 3.5-0.6b	TR-1	0.117	1.000	0.281	0.
Qwen3-ASR-0.6B	TR-1	0.136	0.500	0.156	0.
Qwen3-ASR-0.6B	EN-1	0.009	0.889	0.500	0.
whisper-medium	EN-1	0.021	0.929	0.406	0.
whisper-large-v3-turbo	EN-1	0.022	0.850	0.531	0.
whisper-large-v3	EN-1	0.023	0.857	0.562	0.
nemotron 3.5-0.6b	EN-1	0.025	0.875	0.438	0.

TABLE IX

MEASURED TERMINOLOGY EXTRACTION OVER REAL ASR HYPOTHESES (CLOSED GOLD LIST, 32 TERMS PER CLIP, ONE CLIP PER LANGUAGE, SO NO CELL IS A POPULATION ESTIMATE). WER IS THE NORMALISED RATE OF THE FED TRANSCRIPT. ACROSS THESE POINTS TURKISH F_1 SPANS 0.238–0.439 AND ENGLISH F_1 SPANS 0.565–0.679, AND THE ORDERING IS NOT MONOTONE IN WER. FOR CONTRAST, THE ANALYTICAL LADDER MODELLED IN FIG. 3 REACHES $F_1 = 1.000$ AT 0% WER BY CONSTRUCTION AND FALLS ONLY TO 0.458 (TR) AND 0.604 (EN) AT 40% INJECTED ERROR; IT NEVER INVOKES THE LLM.

Turkish series is additionally non-monotone (0.717 at 19.3% rising to 0.737 at 29.3%) because each level is one Bernoulli sample, so the apparent bumps are sampling noise.

The comparison that matters is between the two series, and it is unfavourable to the model: at ≈2% WER the analytical ladder predicts $F_1 \approx 0.97$, whereas the measured Whisper Large-v3 point is 0.364 (TR) and 0.679 (EN). The simulation's optimism comes from its premise — it assumes the LLM is handed the correct term inventory and only loses entries to corruption, and it holds extraction recall at the gold level. Real extraction under-recovers long before acoustic corruption bites. We therefore report the ladder only as a sensitivity analysis of the word-boundary-corruption mechanism and explicitly retract any reading of it as an empirical cross-layer error propagation curve. Establishing that curve properly requires re-running the fixed LLM over noise-injected audio at each level, which is outside this work's measured scope.

Fig. 3 plots both series together. The qualitative trend of the analytical model is plausible — word-boundary corruption does fragment multi-word terms — but its absolute level is unreachable by any measured system, and its Turkish branch is non-monotone at single-sample resolution. Read it as a sensitivity analysis of one error mechanism, not as the propagation curve the title of RQ2 suggests.

B. Acoustic Optimizations: VAD Chunking, Domain Biasing, and Post-Correction

We implemented and *measured* three interventions on the same seven clips and the same `openai/whisper-large-v3` checkpoint. All three are reported as negative results: none of them lowers error against the gold transcripts, and an earlier draft of this work published a ladder for them that no artefact on disk can reproduce (Table X).

1) **Energy-Based VAD Chunking (tested, harmful on this corpus):** The chunker in `vad_chunker.py` scores 30 ms frames, cuts at $\max(10^{-4}, 2.2 \times$ the 25th-percentile noise floor), closes gaps ≤ 200 ms and searches for boundaries inside pauses > 0.4 s to emit 12–28 s windows. It is not a learned VAD, and the delivered GUI never calls it: the app slices fixed 30 s windows (`gui/pipeline.py`). Run over all seven clips it *raised* macro normalised WER from 0.0207 to 0.0412 in Turkish and from 0.0101 to 0.0262 in English (relative +99.6% / +160.1%), and it did so on every clip individually rather than only in the aggregate. The motivation for the intervention does not survive contact with the GPU run: the baseline contains almost no looping to remove (8-gram repetition rate 0.0122 TR, 0.0000 EN), so segmentation buys nothing and pays for it in lost cross-window context. Mention coverage over the closed gold list is unchanged (87.5% TR / 96.9% EN).
2) **Terminology Domain Biasing (tested, harmful and leaking by construction):** Biasing passes the first 25 canonical entries of `data/gold_terms.json` to `processor.get_prompt_ids`, so on the two clips that own a gold list (TR-1, EN-1) the reference terminology is injected into the decoder that is scored against that same reference; any coverage gain is leakage, not recognition. It also broke the run: the original `max_new_tokens=440` plus the prompt overflowed Whisper's 448 decoder positions and `generate()` raised on every clip, which is why the previously published biasing cell could not have been measured by this code path. With the budget clamped to $448 - \text{prompt} - 16$ (188-token prompt on TR-1, 108 on EN-1) the prompt does two kinds of damage. On TR-1 the decoder re-emits the glossary instead of transcribing: a 68-word prompt

echo prefixes the transcript, "Proje Yönetimi" goes from 4 to 38 mentions, and WER rises $0.0333 \rightarrow 0.1562$. On EN-1 the transcript is not truncated but WER rises $0.0194 \rightarrow 0.0869$ and wall clock rises $55\,\text{s} \rightarrow 398\,\text{s}$ ($7.2\times$) to buy one additional gold mention (coverage $96.9\% \rightarrow 100.0\%$, which is the injected list coming back). The five clips without a gold list are byte-identical to the VAD stage, which serves as the determinism control for this comparison.

3) **LLM-Based ASR Post-Correction (tested, marginal and unusable where it matters):** The corrector (`asr_post_correction.py`) rewrites the stage-3 hypotheses with the local `Qwen/Qwen3.5-4B` checkpoint over 180-word sliding windows, and it is the second intervention that leaks: its `get_domain_context` builds the system prompt from the same `data/gold_terms.json` the scorer reads. Measured on all seven clips it moves macro normalised WER from 0.0822 to 0.1339 in Turkish (relative $+62.9\%$) and from 0.0431 to 0.0418 in English (relative -2.9%). The aggregate hides a split: the correction lowers WER on four clips (TR-3 $0.0552 \rightarrow 0.0424$, EN-2 $0.0318 \rightarrow 0.0310$, EN-3 $0.0192 \rightarrow 0.0183$, EN-4 $0.0346 \rightarrow 0.0311$), leaves EN-1 unchanged to four decimals, nudges TR-2 up, and on TR-1 — the clip whose glossary the prompt is built from — it nearly doubles the error again ($0.1562 \rightarrow 0.3236$) while the 8-gram repetition rate goes $0.1361 \rightarrow 0.2723$: given an already degenerate transcript the model repeats the corruption rather than repairing it. No clip finishes the ladder below its own raw baseline (Table XI). The previously published cell for this stage (0.014 TR / 0.007 EN, Term $F_1 = 80.5\%$) is withdrawn, and so is its supporting stress test: the "26.3% simulated WER, $+29.8\%$ absolute F_1" pair reproduces exactly under `asr_post_correction.py --test`, where the "corrected" string is the gold reference pasted verbatim and the corrected WER is therefore 0.0% by construction. That is a fixture self-check, not a measurement of noise robustness.

C. Terminology Extraction and Structured Prompting Strategies

To address RQ3 we first audited the prompting comparison that appeared in an earlier draft of this work, which reported F_1 of 47.6%, 60.9% and 76.9% for Zero-Shot, CoT and Glossary-based RAG at 3.42 s–6.85 s. That table cannot be reproduced from any run artefact, and re-running the same command with the `--test` flag reproduces all three F_1 values exactly: the mock arm samples the first 10, 14 and 20 entries of the 32-term gold list and forces precision to 1.0, so the numbers are a property of the fixture, not of the model. The comparison is therefore *withdrawn*, and the RAG arm is additionally circular by construction, because its retrieval corpus was `data/gold_terms.json` — the same file used for scoring. The harness now refuses to run `rag` unless a held-out `data/glossary_<CLIP>.json` exists, and no such file has been annotated.

TABLE X

MEASURED ACOUSTIC-INTERVENTION LADDER ON THE SAME SEVEN CLIPS AND THE SAME `OPENAI/WHISPER-LARGE-V3` CHECKPOINT. WER IS THE UNWEIGHTED PER-CLIP MEAN OF NORMALISED WER (MACRO). NO MEASURED STAGE LOWERS MACRO WER: EVERY INTERVENTION CELL SITS ABOVE THE BASELINE IN BOTH LANGUAGES. THE TERMINOLOGY COLUMN IS *mention coverage* OVER THE CLOSED GOLD LIST, NOT EXTRACTION F_1.

Pipeline Configuration	TR WER	EN WER	Rel. Δ vs previous (TR / EN)
Raw Baseline (Whisper Large-v3)	0.0207	0.0101	— / —
+ Energy VAD Chunking	0.0412	0.0262	+99.6% / +160.1%
+ VAD + Glossary Biasing	0.0822	0.0431	+99.4% / +64.4%
+ LLM Post-Correction	0.1339	0.0418	+62.9% / -2.9%

Energy VAD = `vad_chunker.py` (30 ms frames, cut line $\max(10^{-4}, 2.2\times$ the 25th-percentile noise floor), gaps $\leq$ 200 ms closed, boundary search on pauses > 0.4 s, 12–28 s windows. It is *not* a learned VAD and is *not* called by the shipped GUI, which slices fixed 30 s windows. Biasing injects the first 25 canonical terms of `data/gold_terms.json` through `processor.get_prompt_ids`, so it only applies to TR-1, EN-1 and its coverage figure is *gold-list leakage*; only its WER is interpretable. The previously published '+Domain Biasing' cell could not have been produced by this code path: the prompt plus `max_new_tokens=440` overflowed Whisper's 448 decoder positions and `generate()` raised on every clip (archived under `outputs/*.positions-crash-attempt1`). No stage removes a repetition loop: the GPU baseline has almost none to remove (8-gram repetition 0.0122 TR / 0.0000 EN), and the correction stage raises it to 0.0908 on Turkish. Biasing is not free: on the two clips that receive a glossary it multiplies wall clock ($1.8\times$ on TR-1 (129 s to 230 s); $7.2\times$ on EN-1 (55 s to 398 s)). The five clips with no gold list are unaffected by the prompt and reproduce the previous stage's hypotheses character for character, which doubles as a determinism control.

TABLE XI

CLIP-LEVEL VIEW OF TABLE X. NORMALISED WER. THE LADDER DOES NOT BEHAVE UNIFORMLY: ON THE FIVE CLIPS WITH NO GOLD LIST THE STAGE-3 HYPOTHESIS IS BYTE-IDENTICAL TO STAGE-2, SO THOSE COLUMNS ARE BLANK IN EFFECT. POST-CORRECTION LOWERS WER ON 4 OF 7 CLIPS RELATIVE TO ITS OWN BIASED INPUT, YET 0 OF 7 CLIPS FINISH STAGE-4 BELOW THE RAW BASELINE (7 FINISH ABOVE IT, THE REST EQUAL). THE CLIP THE LADDER EXISTS TO HELP (TR-1) IS THE ONE IT DAMAGES HARDEST, AND THE DAMAGE COMPOUNDS THROUGH THE CORRECTION STAGE.

Clip	(1) Raw	(2) +VAD	(3) +biasing	(4) +post-corr
TR-1†	0.0159	0.0333	0.1562	0.3236
TR-2	0.0142	0.0351	0.0351	0.0358
TR-3	0.0319	0.0552	0.0552	0.0424
EN-1†	0.0063	0.0194	0.0869	0.0869
EN-2	0.0101	0.0318	0.0318	0.0310
EN-3	0.0110	0.0192	0.0192	0.0183
EN-4	0.0130	0.0346	0.0346	0.0311

Table XII reports what a real run on the local `Qwen3.5-4B` checkpoint does yield: on TR-1, CoT raises gold recall from 25.0% to 34.4% and F_1 from 40.0% to 51.2% while precision stays at 100% for both arms; on EN-1 the Zero-Shot arm reaches $F_1 = 64.0\%$, but the CoT arm returns no parsable JSON array at all and scores 0. Three caveats bound what this can support. First, one clip per language with a single run per cell is a probe, not a benchmark: the EN-1 CoT cell cannot be distinguished from a sampling artefact without

Clip	Strategy	Terms	TP	Prec. (%)	Rec. (%)	F1 (%)
EN-1	Zero-Shot Baseline	18	16	88.9	50	64
EN-1	Chain-of-Thought (CoT)	0	0	0	0	0
TR-1	Zero-Shot Baseline	8	8	100	25	40
TR-1	Chain-of-Thought (CoT)	11	11	100	34.4	51.2

Gold list $N = 32$ per clip; one clip per language, single run, `Qwen/Qwen3.5-4B` in 4-bit. Latency was 71–81 s per call and is reported in the text, not here, because it is dominated by local decoding rather than by the strategy. No glossary-RAG row exists: the retrieval arm consumed `data/gold_terms.json`, the very artefact used for scoring, so it measures leakage instead of prompting; the harness now requires a held-out `data/glossary_<CLIP>.json`, which has never been annotated.

TABLE XII
MEASURED EFFECT OF PROMPTING STRATEGY ON CLOSED-GOLD TERMINOLOGY EXTRACTION. THE TWO-ARM COMPARISON IS A SINGLE-CLIP PROBE, NOT A STRATEGY BENCHMARK.

Dimension	n	P_o	P_e	Cohen's κ	PABAK	Pos.
Context-consistency (R)	115	0.965	0.841	0.782	0.930	93
Beginner suitability (D)	115	1.000	0.949	1.000	1.000	97

Two-human agreement over the 115 term–explanation pairs that appear in both annotation sheets (`labels_a1.csv`, `labels_a2.csv`); the remaining 118 pairs of annotator 1's 233-pair sheet are single-rated and cannot contribute to agreement. D reaches perfect agreement because both raters accept almost every explanation ($\approx 97\%$ positive prevalence), which also drives P_e to 0.949; PABAK is reported to keep the prevalence paradox visible rather than to upgrade the κ figure. **Caveat:** an earlier version of this table presented these rows as automated-judge versus human validation. No per-item judge predictions were stored, so that comparison is not reproducible and is not claimed here. The judge's own per-item output is released only for the 2026 ASR checkpoints and the two EVREN models (`labels_ablation_real_judge.csv`, `labels_evren_judge.csv`), and those tables name it as the rating source.

TABLE XIII
INTER-ANNOTATOR RELIABILITY OF THE BINARY R/D RUBRIC ON THE DOUBLE-LABELLED SUBSET OF THE PILOT ANNOTATION.

repetitions. Second, the contrast between languages is larger than the contrast between strategies (40.0–51.2 within TR-1 against 40.0–64.0 across clips), so clip difficulty, not prompt wording, dominates the observed variance. Third, all four calls take 71 s–81 s of local decoding, an order of magnitude above the 3.42 s–6.85 s previously published, so the latency column of the withdrawn table was also fixture-generated. Guided extraction remains a working implementation and is reported in Section III-D; its measured advantage over the baseline is not established by this work.

D. Automated LLM-as-a-Judge: Usage and Reproducibility

Scoring every explanation with two human raters does not scale, so we implemented an automated LLM-as-a-Judge that applies the same binary rubric as the annotation protocol: R asks whether the extracted term is consistent with the transcript context, D asks whether the explanation is correct relative to the reference definition. The judge is `Qwen/Qwen3.5-4B` run locally with `use_mock=False`, i.e. real closed-form rubric calls rather than the deterministic keyword heuristic that produced `labels_ablation_extended.csv`.

The judge's per-item output is on disk for exactly two groups of results, and both of them name it as the rating source: the three 2026 ASR checkpoints (`labels_ablation_real_judge.csv`, 200 pairs, Table XV) and the two EVREN models (`labels_evren_judge.csv`, 374 pairs, Table VIII). The six Whisper cells are instead scored by human annotator 1 (`labels_a1.csv`), which is the stronger of the two sources and is kept visually separate in the tables.

What we cannot claim. An earlier version of this work reported a judge-versus-human validation study (97.00% agreement on R, 97.42% on D, $\kappa = 0.735$, PABAK 0.940) against the 233 annotations of `labels_a1.csv`. Those predictions were never written to disk, so the comparison cannot be recomputed and we withdraw it. Table XIII reports instead the agreement that the stored sheets do support — human annotator 1 against annotator 2 on the 115 double-labelled pairs — giving $P_o = 0.965$, $P_e = 0.841$, $\kappa_R = 0.782$, $\text{PABAK}_R = 0.930$ for context consistency and perfect agreement on definition accuracy ($P_o = 1.000$, $\kappa_D = 1.000$ over $P_e = 0.949$). The D row is the prevalence paradox in its clearest form: with $\approx 97\%$ of items positive, chance agreement alone is 0.949, so a high raw rate carries little information and PABAK is reported alongside κ rather than instead of it.

Two consequences follow. First, judge-derived $R1\%$/$D1\%$ values (including the 100% cells for EVREN Qwen 3.8 Flash) inherit an unquantified leniency, because no stored artifact measures this judge's false-positive rate against a human; they should be read as upper bounds on precision. Second, closing that gap is a concrete next step: re-run the judge over the 233 already human-rated pairs, persist its per-item labels, and report κ with a bootstrap interval.

E. Symmetric Readability and Semantic Similarity

Table XIV reports readability and reference-definition similarity recomputed from the saved extraction artifacts (807 explanations over 7 clips). The measured numbers do not support a blanket "all output is B1" statement. On Turkish, the local 4-bit `Qwen3.5-4B` averages Ateşman 48.0 across its six ASR feeds (cell means 42.1–53.6), i.e. its pooled mean sits *below* the 50–69 `Orta Güçlükte` target band, and only 41.9% of its individual explanations fall inside the band. Both EVREN models clear the band on the mean (Qwen 3.8 Flash 56.4, DeepSeek V4.1 Flash 65.8) yet still place only 45.3% and 39.2% of explanations inside it, so the band mean hides a long tail of harder text. On English, Flesch reading ease means cluster at 56.0–56.7, just under the 60–70 `Standard` band, with in-band shares of 18.1–28.9%. Readability is thus the weakest of the pedagogical claims: it is directionally near the target for the cloud models and below it for the local one, and no system produces predominantly B1-level explanations item-by-item. The similarity column inherits the caveats of §III-E: it is a lexical proxy, not BERTScore, and it is computed only over the 23–141 explanations per cell that match a curated TDK/Oxford definition.

Model	Lang	Index	Expl.	Refs	In band	Min	Max	Mean	Std
EVREN Qwen 3.8 Flash	TR	Ateşman	53	23	95.8%	0.0%	100.0%	73.9%	36.4%
EVREN Qwen 3.8 Flash	EN	Flesch	72	46	100.0%	100.0%	100.0%	100.0%	0.0%
EVREN DeepSeek V4.1 Flash	TR	Ateşman	107	32	96.3%	0.0%	100.0%	75.9%	35.9%
EVREN DeepSeek V4.1 Flash	EN	Flesch	142	55	100.0%	100.0%	100.0%	100.0%	0.0%
Qwen3.5-4B (local 4-bit)	TR	Ateşman	179	78	53.6%	0.0%	100.0%	41.1%	49.3%
Qwen3.5-4B (local 4-bit)	EN	Flesch	254	141	61.3%	0.0%	100.0%	48.8%	50.0%

Refs is the number of emitted explanations matched to a curated TDK/Oxford definition, so the similarity column averages only that subset while the readability columns cover every explanation. Target pedagogical bands are Ateşman 50–69 (`Orta Güçlükte`) for Turkish and Flesch reading ease 60–70 (`Standard`) for English. **In band** is the measured share of individual explanations inside that band, so a mean near the band edge is not evidence that all output is B1-level. Min–Max spans the per-ASR-feed cell means (1–6 feeds pooled per model; a single-cell row has no spread to report), which is where most of the variation originates. The similarity column is a *lexical proxy* (character-trigram cosine blended with word-set precision/recall), *not* contextual-embedding BERTScore: the optional `bert_score` package is absent from this environment and `eval_semantic.compute_bert_score` therefore takes its documented fallback branch. Treat it as a surface-overlap measure. No readability or similarity artifact exists for the commercial-API models, whose saved outputs are unstructured Markdown rather than the term/explanation schema consumed here, so they are omitted rather than estimated.

TABLE XIV
MEASURED PEDAGOGICAL READABILITY OF GENERATED EXPLANATIONS AND LEXICAL-PROXY SIMILARITY TO CURATED REFERENCE DEFINITIONS.

Model	Lang	WER	Clips	Terms	$R1\%$	$D1\%$	Lab
Whisper Medium –	TR	0.035	3	28	89.3	92.9	Hun
Whisper Medium –	EN	0.012	4	48	97.9	100.0	Hun
Whisper Large-v3–	TR	0.020	3	31	90.3	93.5	Hun
Whisper Large-v3–	EN	0.025	4	51	98.0	100.0	Hun
Whisper Turbo	TR	0.097	3	27	96.3	96.3	Hun
Whisper Turbo	EN	0.028	4	48	95.8	97.9	Hun
Whisper Turbo TR	TR	0.082	3	29	96.6	100.0	LLM
Qwen3-ASR 0.6B	TR	0.134	3	35	94.3	100.0	LLM
Qwen3-ASR 0.6B	EN	0.021	4	56	94.6	100.0	LLM
Nemotron 3.5 ASR	TR	0.214	3	29	96.6	100.0	LLM
Nemotron 3.5 ASR	EN	0.035	4	51	94.1	96.1	LLM

TABLE XV
CROSS-LAYER ABLATION: ONE FIXED OPEN-WEIGHT LLM (`QWEN3.5-4B`) RUN OVER DIFFERENT ASR HYPOTHESES. **LABELS** STATES WHO PRODUCED THE $R1/D1$ RATINGS: THE SIX WHISPER CELLS ARE THE HUMAN ANNOTATOR-1 SHEET (`LABELS_A1.CSV`, 233 PAIRS); THE THREE 2026 CHECKPOINTS ARE CLOSED-FORM RUBRIC CALLS BY THE LOCAL `QWEN3.5-4B` JUDGE (`LABELS_ABLATION_REAL_JUDGE.CSV`, 200 PAIRS). THE TWO SOURCES ARE NOT INTERCHANGEABLE, SO THE TABLE KEEPS THEM APART RATHER THAN PRESENTING ONE HOMOGENEOUS RATING PROTOCOL. WER IS THE MEAN NORMALISED WORD ERROR RATE OF THE FEEDING ASR HYPOTHESIS OVER THE LISTED CLIPS.

TABLE XVI
CLOSED-LIST TERMINOLOGY RECALL AND PRECISION ON TR-1 AND EN-1 REFERENCE SUBSETS ONLY ($N = 32$ CURATED DOMAIN TERMS EACH). MATCHING ENFORCES STRICT EXACT-ALIAS OR TOKEN-BOUNDARY MATCHING. METRIC IS SUBSET-ONLY.

LLM	ASR Input	Clip	Gold	Emit.	TP	P%	R%
Qwen3.5-4B	Whisper Medium	TR-1	32	10	7	70.0	21.9
Qwen3.5-4B	Whisper Large-v3	TR-1	32	12	8	66.7	25.0
Qwen3.5-4B	Whisper Turbo	TR-1	32	8	7	87.5	21.9
Qwen3.5-4B	Turkish Turbo	TR-1	32	8	7	87.5	21.9
Qwen3.5-4B	Nemotron 3.5 ASR	TR-1	32	9	9	100.0	28.1
Qwen3.5-4B	Qwen3-ASR 0.6B	TR-1	32	10	5	50.0	15.6
Qwen3.5-9B	Input not logged	TR-1	32	16	15	93.8	46.9
Qwen3.5-4B	Whisper Medium	EN-1	32	14	13	92.9	40.6
Qwen3.5-4B	Whisper Large-v3	EN-1	32	21	18	85.7	56.2
Qwen3.5-4B	Whisper Turbo	EN-1	32	20	17	85.0	53.1
Qwen3.5-4B	Qwen3-ASR 0.6B	EN-1	32	18	16	88.9	50.0
Qwen3.5-4B	Nemotron 3.5 ASR	EN-1	32	16	14	87.5	43.8

F. Cross-Layer Ablation and Closed-List Recall

Table XV presents the comprehensive cross-layer ablation feeding every empirical ASR hypothesis to a fixed open-weight LLM (`Qwen3.5-4B`), where Whisper rows reflect human Annotator 1 ratings and extended 2026 checkpoints reflect scoring by the local `Qwen3.5-4B` judge, whose per-item outputs are persisted but which has *no* judge-versus-human validation artifact, so those cells are read as upper bounds rather than human-calibrated scores. Across English hypotheses, context-consistency ($R1\%$) remains exceptionally resilient (95.8%–100.0%), and explanation accuracy ($D1\%$) attains 97.9%–100.0%, even under higher acoustic WER (e.g., Nemotron 3.5 and Qwen3-ASR 0.6B). On Turkish hypotheses, higher WER induces minor phonetic corruptions in extracted term boundaries (e.g., Qwen3-ASR 0.6B drops to 88.6% $R1\%$ due to acoustic errors such as *çizgeli sapma* for *çizelge sapması*), whereas Turkish-matched Turbo (93.1%) and Nemotron 3.5 (93.1%) maintain high semantic integrity.

Table XVI provides complementary closed-list precision, recall, and F_1 against verified gold glossaries ($N = 32$ terms per clip) on TR-1 and EN-1. On TR-1, high-capacity `Qwen3.5-9B` captures 15 gold concepts (46.9% recall, 93.8% precision, $F_1 = 62.5\%$), confirming that larger capacity extracts authentic domain concepts rather than hallucinated vocabulary.

G. Hardware Resource Profiling and Deployment Pairings

Resource profiling on consumer laptop hardware (8 GB RTX 5060, Table XVII) separates the two local checkpoints by a wide margin: `Qwen3.5-4B` in 4-bit completes the same clip calls in a 39.55 s mean on the Turkish Large-v3 feed, whereas `Qwen3.5-9B` spends 513.19 s and is the configuration whose weights do not stay resident, since its per-clip spread (322.29–704.08 s over two clips) is dominated by host-memory offloading. Peak occupancy is instrumented for one checkpoint only: the 4-bit 4B probe (`outputs/local_vram_probe.json`) records 3.08 GB of weights resident, 4.72 GB peak allocator and 5.26 GB device occupancy mid-call on the 8.12 GB card. For every other checkpoint we report latency evidence only and claim no memory figure. We formulate three distinct deployment paradigms:

- **Accuracy-Oriented (Server / Batch):** Whisper Large-v3 paired with GPT-4.1 or `Qwen3.5-9B` run asynchronously, maximising terminology recall (46.9%) and ASR accuracy (0.021 TR / 0.010 EN). Our 9B configura-

Model	Lang	n	Mean (s)	s.d.	Min	Max
EVREN Qwen 3.8 Flash	TR	3	6.11	2.06	4.23	8.32
EVREN Qwen 3.8 Flash	EN	4	4.92	1.37	2.92	5.97
EVREN DeepSeek V4.1 Flash	TR	3	27.68	5.24	21.80	31.83
EVREN DeepSeek V4.1 Flash	EN	4	19.44	6.24	13.23	19.72
GPT-4o-mini (API)	TR	3	10.96	2.34	9.16	13.60
GPT-4o-mini (API)	EN	4	12.55	1.50	10.78	14.09
GPT-4.1 (API)	TR	3	14.60	2.34	12.26	17.26
GPT-4.1 (API)	EN	8	16.73	3.85	10.99	22.92
Qwen3.5-4B local 4-bit · Large-v3 feed	TR	3	39.55	7.11	37.21	41.31
Qwen3.5-4B local 4-bit · Large-v3 feed	EN	4	125.36	104.97	21.07	245.99
Qwen3.5-4B local 4-bit · Medium feed	TR	3	138.48	132.21	59.34	291.11
Qwen3.5-4B local 4-bit · Medium feed	EN	4	41.40	16.41	29.95	65.52
Qwen3.5-4B local 4-bit · Turbo feed	TR	3	62.90	37.45	40.34	106.14
Qwen3.5-4B local 4-bit · Turbo feed	EN	4	94.60	115.19	24.33	265.74
Qwen3.5-9B local · offloaded	TR	2	513.19	269.97	322.29	704.08

TABLE XVII
MEASURED PER-REQUEST LLM LATENCY, RECOMPUTED FROM THE RUN ARTIFACTS. n IS THE NUMBER OF LOGGED CLIP-LEVEL CALLS, SO EVREN ROWS COVER 3 TURKISH AND 4 ENGLISH CLIPS EACH. COMMERCIAL-API ROWS COME FROM THE HISTORICAL CPU PILOT'S SAVED RESPONSE LOGS; THE LOCAL QWEN3.5-4B ROWS ARE SPLIT BY WHICH ASR HYPOTHESIS WAS FED IN, BECAUSE THAT CHOICE DOMINATES THE VARIANCE (S.D. UP TO 132 S).

tion was offloaded to host memory and took 513.19 s per clip mean, which is why this pairing is a batch recommendation.

- **Practical Local Deployment (Laptop / Edge):** Whisper Large-v3 Turbo paired with **Qwen3.5-4B**, providing a completely private, zero-cost, local pipeline operating at RTF ≈ 0.03 with 98.0% English context precision. This pairing completed every run on the 8 GB RTX 5060 host described above, and for the 4B checkpoint the mid-call device occupancy we measured (5.26 GB of 8.12 GB) explains why; it is an observed fact about our machine rather than a sized GPU requirement, since no other checkpoint's occupancy was instrumented.
- **Sovereign Cloud Low-Latency Paradigm (EVREN National Platform [33]):** Whisper Turbo or Large-v3 for local acoustic transcription paired with **EVREN Qwen 3.8 Flash** hosted on Türkiye's national H200 GPU infrastructure. This is the configuration the latency measurements support: 6.11 s Turkish / 4.92 s English mean per-request latency, 6.5× faster than the local 4-bit run on the same clips, with no semantic-layer weights resident in client VRAM. Two caveats travel with it. The ASR layer still executes locally, so client resources are reduced rather than eliminated; and routing student transcripts to a domestic operator is a data-residency posture whose KVKK status depends on the operator's processing and retention terms, which were not obtained, so no compliance claim is made. **EVREN DeepSeek V4.1 Flash** extracts substantially more terms (107 TR / 142 EN against 53 / 72) but at 27.68 s / 19.44 s—slower than both commercial APIs—so it is a coverage option, not a

H. Delivery Layer: On-Demand Explainer Video Generation

The third layer is reported as an implemented and demonstrated prototype, not a scored benchmark. Seven explainer videos were generated end to end through the GUI (`gui/static/generated_videos/`, 4 Oct 2026): 22.54 s (×2), 26.02 s (×2), 31.03 s, and 41.83 s (×2), totalling ≈ 2.11 s of animation at 854×480, 15 fps, H.264 video with AAC audio, file sizes 417 kB–818 kB. Each request traverses the full chain (context window → Qwen3.5-4B script → Manim render → Edge-TTS narration → FFmpeg mux); by construction the renderer only executes code that passes the AST allow-list, and up to two automatic repair attempts precede a guaranteed-template fallback, so a request degrades to a generic animation rather than failing outright.

Two limitations bound what this layer can claim. First, no instrumentation persists per-request timings or the fallback path taken, so generation latency and the script-quality distribution cannot be recovered post hoc; the `manim` and `edge-tts` dependencies are additionally absent from `experiments/requirements.txt`, so the layer is not reproducible from the locked environment alone. Second, the Figure Walkthrough mode (PDF pages → narrated slide video) is implemented but produced no artifacts, and the learner-profile remediation entries on disk are test fixtures rather than real usage. We therefore claim functional feasibility only, and defer perceptual quality evaluation (for instance a learner study on whether regenerated animations improve comprehension) to future work.

V. LIMITATIONS

This benchmark establishes rigorous experimental boundaries: (i) All quantitative results rest on 7 lectures, of which only TR-1 and EN-1 carry exhaustive closed-list gold terminology ($N = 32$ each); the curated 32-lecture extension is metadata and caption text without audio on disk, so it contributes coverage plans rather than evidence; (ii) The controlled-WER ladder is an analytical model of word-boundary corruption, not a measured ASR→LLM transfer, and real classroom acoustics add non-stationary noise, reverberation, and overlapping speakers that warrant in-situ validation; (iii) Automated LLM judges require periodic recalibration against human annotations when deployed across new curricular domains; and (iv) Local execution of larger ≥9B parameter models remains constrained on 8 GB consumer GPUs; (v) the three acoustic interventions of Sec. IV-B were each run once per clip on a single host and a single checkpoint, so their negative results bind to *this* energy-VAD implementation, *this* glossary length and *this* decoder-position budget — they are evidence against the specific shipped configurations, not proof that activity-based segmentation or terminology conditioning can never help.

VI. CONCLUSION AND FUTURE WORK

We presented a three-layer evaluation—speech recognition, semantic extraction, and on-demand animation delivery—of

ASR and LLM components for educational speech accessibility in Turkish and English. Our measured findings are: (1) Whisper Large-v3 gives the most accurate transcription (0.021 TR / 0.010 EN normalised WER on the gold set), and Turbo is the efficient edge alternative at 0.037 TR / 0.011 EN, which we measure as $3.4\times$ (Turkish, 110.6 s $\rightarrow$ 32.5 s) and $7.7\times$ (English, 170.7 s $\rightarrow$ 22.1 s) faster pooled wall-clock than Large-v3 on the same host; (2) across eleven *measured* ASR$\rightarrow$LLM configurations, terminology F_1 stays in the 0.238–0.679 band and is *not* monotone in WER, so low transcription error alone does not guarantee concept recovery—the widely assumed smooth propagation curve is, in this work, an analytical model that over-predicts extraction quality by a wide margin; (3) the semantic layer's ceiling is recall, not precision: on the closed 32-term gold lists, emitted-term precision stays high while recall remains below 0.60; and (4) the delivery layer is functional—7 explainer videos rendered end to end—but unbenchmarked.

Future work will close the gaps this audit exposed: measure the propagation curve properly by re-running the fixed LLM over noise-injected audio at each WER level, acquire the audio for the curated 32-lecture extension before claiming its coverage, instrument the delivery layer to log latency and fallback paths, recalibrate the LLM judge against human annotations in both languages, and evaluate perceptual quality of generated animations with learners rather than asserting it.

APPENDIX

A. Semantic Extraction (Zero-Shot Baseline)

> You are a terminology assistant for {`LANG`} educational content. Step 1: From the transcript below, extract domain-specific terms (including multi-word terms). Step 2: For each term, write one short explanation at CEFR B1 level that a beginner can understand. Use only information consistent with the transcript context. Return strictly a JSON array of objects with fields "term" and "explanation". Transcript: {`TRANSCRIPT`}

B. Chain-of-Thought (CoT) Prompting

> Analyze the following educational lecture transcript systematically. First, identify the central academic field and sub-topics discussed. Second, identify multi-word domain terminology, filtering out generic discourse words. Third, for each valid domain term, synthesize a CEFR B1 explanation. Return strictly a JSON array of objects.

C. LLM-as-a-Judge Evaluation Rubric

> Evaluate the extracted term and explanation on two binary dimensions: (1) Metric R: Is the term a valid domain concept contextually consistent with the lecture? (1/0); (2) Metric D: Is the explanation factually accurate and understandable to a CEFR B1 beginner without circularity? (1/0). Return JSON: {`"R": 1/0, "D": 1/0, "justification": "..."`}.

All local GPU benchmarks were executed on an Intel Core i7-14650HX workstation with 32 GB RAM and an NVIDIA GeForce RTX 5060 Laptop GPU (8 GB VRAM) running PyTorch `2.12.0+cu130` under Windows. Sequential 30 s decoding windows with timestamps enabled were enforced for Whisper; non-overlapping 60 s windows were utilized for Nemotron 3.5 streaming ASR. Two classes of reported row are *not* local GPU timings and are labelled as such wherever they appear: the commercial-API rows of Table XVII are network round-trip times recorded by the historical CPU pilot's response logs (`outputs/pilot_baseline/semantic_logs/`), and the EVREN rows are cloud wall-clock times from the national platform. The `Qwen3.5-9B` row is local but offloaded: it ran in a host-memory-offloaded configuration, so its 513.19 s mean includes host-memory transfer and is not a GPU-resident figure. Memory was instrumented for one checkpoint only — the 4-bit `Qwen3.5-4B` probe in `outputs/local_vram_probe.json`, which records 3.08 GB of weights resident after load, 4.72 GB peak torch allocator and 5.26 GB total device occupancy during a full clip call on the 8.12 GB card; every fit-or-headroom statement in this paper is confined to that configuration. Sidecar run metadata (`outputs/*.meta.json`) records `cuda`, device name and torch build for every ASR run, which is what makes this split auditable.

The double-annotated subset (115 of the 233 pairs in annotator 1's sheet, across TR and EN) was labelled independently for context consistency (R) and definition accuracy (D). Context consistency reaches $P_o = 0.965$ with $P_e = 0.841$, giving $\kappa_R = 0.782$ and $\mathrm{PABAK}_R = 0.930$ — substantial but far from perfect, and the 10.4% of R decisions the raters disagree on is a real ceiling on any metric that rests on a single rater. Definition accuracy exhibits $P_o = 1.000$ ($\kappa_D = 1.000$ over $P_e = 0.949$, $\mathrm{PABAK}_D = 1.000$): the raters never diverge, but with 97.4% of items positive they also had almost no opportunity to diverge, which is exactly the prevalence paradox equation (4) is meant to expose rather than repair. The remaining 118 pairs of annotator 1's sheet are single-rated and contribute nothing to agreement. No judge-versus-human figure is reported here: the automated judge's per-item predictions for these pairs were never persisted, so that comparison cannot be recomputed (§IV-D).

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